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Paper Code : 21304
F-404
B.C.A. (First Semester)
Examination, 2024-25
(New Course)

Paper - BCA-104-N

MATHEMATICS - I

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Show that $\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x^3}}$ does not exist.

   (b) A function f(x) is defined as:

$$f(x)=\begin{cases}\frac{x^2}{a}-a & : x<a\\ 0 & : x=a\\ a-\frac{a^2}{x} & : x>a\end{cases}$$

Show that the function f(x) is continuous at x=a.

2. (a) If $x=\sin\left\{\frac{(\log y)}{a}\right\}$, prove that:

   $(1-x^2)y_{n+2}-(2n+1)y_{n+1}x-(n^2+a^2)y_n=0$

   (b) Evaluate $\int_0^{\pi/2}\log\sin x\,dx$.

3. (a) Solve :

   $(D^2+D+1)y=\sin 2x$

   (b) Evaluate:

$$\frac{xdx+ydy}{xdy-ydx}=\sqrt{\frac{a^2-x^2-y^2}{x^2+y^2}}$$

4. (a) Obtain the reduction formula for $\int \sec^n x\,dx$.

   (b) Find n^{th} derivative of $\tan^{-1}\frac{x}{a}$ with respect to x.

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5. (a) Show that the function f defined as-

$$f(x) = \begin{cases} 0 & ; \quad x = 0 \\ \frac{1}{2} - x & ; \quad 0 < x < \frac{1}{2} \\ \frac{1}{2} & ; \quad x = \frac{1}{2} \\ \frac{3}{2} - x & ; \quad \frac{1}{2} < x < 1 \\ 1 & ; \quad x = 1 \end{cases}$$

has 3 points of discontinuity and also find them. https://www.mjpruonline.com

(b) State and prove Jacobian theorem.

6. (a) If $u = \tan^{-1} \frac{xy}{\sqrt{1 + x^2 + y^2}}$ show that:

$$\frac{\partial^2 w}{\partial x \partial y} = \frac{1}{(1 + x^2 + y^2)^{3/2}}$$

(b) Show that the vectors $\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k}, \vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} - 4\hat{k}$ form a right angled triangle.

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7. (a) Find the eccentricity and co-ordinate of the foci of the curve $3x^2 - y^2 = 4$

(b) Find the center, vertices, foci eccentricity and latus rectum of the hyperbola.

$9x^2 - 16y^2 - 18x - 64y - 199 = 0$

8. (a) Show that:

curl (grad r^n) = 0

(b) Prove that:

$$\nabla^2 (r^n \vec{r}) = n(n + 3) r^{n-2} \vec{r}$$

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: 21304

F-404

B.C.A. (First Semester)

Examination, 2023-24

(New Course)

Paper - BCA-104-N

MATHEMATICS - 1

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Compute $\lim_{x\to 0}\frac{e^{\sin x}-1}{x}$

(b) Show that $f(x) = \log x$, $x\in(0,\infty)$ is continuous at every point of its domain.

2. (a) Find the nth derivative of $y=\tan^{-1}(x/a)$.

(b) If $y=\cos(m\sin^{-1}x)$, show that

$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}+(m^2-n^2)y_n=0$

3. **(a)** If $x^y y^x=c$, show that at $x=y=z$

$\frac{\partial^2 z}{\partial x\partial y} = -(x\log ex)^{-1}$

(b) If $u=xy+yz+zx$, $v=x^2+y^2+z^2$, and $w=x-y+z$, show that they are functionally related and find the relation between them.

4. **(a)** Evaluate $\int_0^{\pi/6}\cos^3 3\theta\sin^3 6\theta d\theta$

(b) Find the perimeter of the circle $x^2+y^2=a^2$.

5. **(a)** Evaluate $\int_0^1\int_0^{x^2}(x^2+y^2)dx\,dy$

(b) Evaluate $\int_0^a\int_0^{a-x}\int_0^{a-x-y}x^2\,dx\,dy\,dz$

6. (a) Solve $(x+1)\frac{dy}{dx} - y = e^x(x+1)^2$

(b) Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = e^{2x} + \cos x$

7. (a) A particle moves along the curve $x=t^3+1$, $y=t^2$, $z=2t+5$, where t is the time. Find the components of its velocity and acceleration at t=1 in the direction $2i+3j+6k$.

(b) If $\vec{a} = x^2yz\,\hat{i} - 2xz^3\hat{j} + xz^2\hat{k}$ and $\vec{b} = 2z\,\hat{i} + y\hat{j} - x^2\hat{k}$, find the value of $\frac{\partial^2}{\partial x^2}(\vec{a}\times\vec{b})$ at the point (1,0,1).

8. (a) Find the value of n for which the vector $r^n\vec{r}$ is solenoidal, where $\vec{r} = x\,\hat{i} + y\hat{j} + z\hat{k}$

(b) Prove that, for every field $\vec{v}$; div curl $\vec{v} = 0$.

9. (a) The cost of fencing a circular field at the rate of Rs. 24 per metre is Rs. 5280. The field is to be ploughed at the rate of Rs. 0.50 per m^2. Find the cost of ploughing the field ($\pi = 22/7$).

(b) Find the equation of a line making an angle of 150° with the x-axis and cutting off an intercept 2 from y-axis.

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F - 404

B. C. A. (First Semester) Examination, 2022-23

(New Course)

Paper No. BCA-104-N

MATHEMATICS-I

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Show that the function f defined as:

$$f(x)\begin{cases} 0 & ; \quad x = 0 \\ \frac{1}{2} - x & ; 0 < x < \frac{1}{2} \\ \frac{1}{2} & ; \quad x = \frac{1}{2} \\ \frac{3}{2} - x & ; \frac{1}{2} < x < 1 \\ 1 & ; \quad x = 1 \end{cases}$$

has 3 points of discontinuity and also find them.

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(b) Find the whole area of the curve $a^2y^2=x^3(2a-x)$

2. (a) If $u = \tan^{-1}\frac{xy}{\sqrt{1+x^2+y^2}}$ show that : $\frac{\partial^2 u}{\partial x \partial y} = \frac{1}{(1+x^2+y^2)^{3/2}}$

(b) State and prove jacobian theorem.

3. (a) Evaluate : $\int_0^{2a} \frac{x^{a/2}}{\sqrt{2a-x}}dx$

(b) Prove that : $\nabla^2(r^n\vec{r}) = n(n+3)r^{n-2}\vec{r}$

4. (a) Evaluate : $\lim_{x\to 0} \frac{x^{1/2}\tan x}{(e^x-1)^{1/2}}$

(b) Solve : $(D^2+Dy=8x^2e^{2x} \sin 2x$

5. (a) Find the eccentricity and co-ordinate of the foci of the curve $3x^2-y^2=4$

(b) Find the center, vertices, foci, eccentricity and latus rectum of the hyperbola $9x^2-16y^2-18x-64y-199 = 0$

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6. (a) Find nth derivative of $\tan^{-1}\frac{x}{a}$ with respect to x.

(b) If $I_n = \int_0^{\pi/3} \tan^n x\,dx$, show that

$(n-1)(I_n + I_{n-2}) = (\sqrt{3})^{n-1}$

7. (a) Show that the vectors $\bar{a} = 3\hat{i} - 2\hat{j} + \hat{k}, \hat{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\hat{c} = 2\hat{i} + \hat{j} - 4\hat{k}$ form a right angled triangle.

(b) Obtain the reduction formula for

$\int \sec^n x\,dx$

8. (a) If $x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

Show that : $\frac{\partial(x,y,z)}{\partial(r,\theta\phi)} = r^2\sin\theta$

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(b) Determine the value of a,b,c for which the function:

$$f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x}; & x < 0 \\ c; & x = 0 \\ \frac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}}; & x > 0 \end{cases}$$

is continuous at x=0

9. (a) Show that the volume of a sphere of radius 'a' is $\frac{4}{3}\pi a^3$

(b) solve : $(D^4+m^4)y = 0$

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F-404

B.C.A. (First Semester)

Examination, 2021-22

(New Course)

Paper-No. BCA-104-N

MATHEMATICS-I

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Show that the limit of a product is equal to the product of limits.

   (b) Evaluate the following limits, if they exist :

   (i) $\lim_{x\to 0} \sin\frac{1}{x}$

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(ii) $\lim_{x\to 0} \frac{1}{x}.e^{1/x}$

2. A function f(x) is defined as follows :

$$f(x) = \begin{cases} (x^2/a)-a & ; \ x < a \\ 0 & ; \ x = a \\ a-(a^2/x) & ; \ x > a \end{cases}$$

Prove that the function f(x) is continuous at x=a.

3. (a) Explain the types of discontinuity.

   (b) Show that the function :

$$f(x) = \begin{cases} 1+x; & \text{if } x \le 2 \\ 5-x; & \text{if } x > 2 \end{cases}$$

continuous at x=2.

4. (a) Find the nth derivative of y=(ax+b)m.

   (b) (i) If $y^{1/m}+y^{-1/m}=2x$, prove that

$$(x^2-1)y_{n+2}+(2n+1)xy_{n+1}+(n^2-m^2)y_n=0.$$

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(ii) If $u=\log(x^3+y^3+z^3-3xyz)$, show that

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}.$$

5. (a) Solve : $\frac{dy}{dx} = \frac{2x+2y-2}{3x+y-5}$.

(b) Solve : $(D^2+1)\,y = \sin x \sin 2x$.

6. (a) Evaluate : $\int_0^{\pi/4} \sqrt{\tan\theta}\, d\theta$.

(b) Write reduction formula for $\int x \sin^n x\, dx$.

7. (a) If $u=x+y+z$, $v=x^2+y^2+z^2$ and $w=yz+zx+xy$, prove that:

(grad u). [(grad v)×(grad w)]=0.

(b) Prove that a. {∇(v.a)–∇x(v×a)}=div v;

where 'a' is a constant unit vector.

8. (a) Solve : $\frac{d^2y}{dx^2} = \sec ax$

(b) Solve : $(D^2-2D+1)y=x^2e^{3x}$.

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9. (a) Find the equation of the ellipse whose eccentricity is $\frac{4}{5}$ and axis are along the co-ordinate axis and with foci at $(0, \pm 4)$.

(b) A triangle ABC has vertices at points A=(8, 1), B=(–1, 4) and C=(6, –3). Determine the measure of the centre and radius of a circumscribed circle.

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Paper Code : 21304

F-404

B.C.A. (First Semester) Examination, 2021

(New Course)

Paper-No. BCA-104-N

MATHEMATICS-I

Time : Three Hours | *Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) If $x=\sin t$, $y=\sin p\,t$, prove that
$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + P^2y = 0.$$ 7

(b) If $y=e^{a\sin^{-1}x}$, prove that $(1-x^2)\,y_{n+2}-(2n+1)\,xy_{n+1}-(n^2+a^2)\,y_n=0$. Hence find the value of y_n when $x=0$. 7

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2. (a) If $\frac{x^2}{a^2+u}+\frac{y^2}{b^2+u}+\frac{z^2}{c^2+u}=1$ prove that
$$\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial u}{\partial z}\right)^2=2\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}\right).$$ 7

(b) If $u=\sin^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$, prove that
$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{2}\tan u.$$ and
$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=-\frac{\sin u\cos 2u}{4\cos^3 u}.$$ 7

3. (a) If $x+y=2e^{\theta}\cos\phi$ and $x-y=2ie^{\theta}\sin\phi$, show that $\frac{\partial^2 u}{\partial\theta^2}+\frac{\partial^2 u}{\partial\phi^2}=4xy\frac{\partial^2 u}{\partial x\,\partial y}$. 7

(b) If $u^3+v^3=x+y$ and $u^2v^2=x^3+y^3$, show that
$$\frac{\partial(u,v)}{\partial(x,y)}=\frac{1}{2}\frac{y^2-x^2}{uv(u-v)}.$$ 7

4. (a) Integrate : 7
$$\int \sin^4 x\cos^2 x\,dx.$$

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(b) Evaluate : 7

$\int_0^{\pi/2} \log \sin x \, dx$.

5. (a) Find the volume of the solid obtained by revolving the cissoid $y^2(2a-x)=x^3$ about its asymptote. 7

(b) Prove that $\text{div}(r^n R)=(n+3)r^n$. Hence show that R/r^3 is solenoidal. 7

6. (a) Show that $\nabla^2(r^n)=n(n+1)r^{n-2}$. 7

(b) Solve $3x(1-x^2)y^2\frac{dy}{dx}+(2x^2-1)y^3=ax^3$. 7

7. (a) Solve, by changing the independent variable, $x\frac{d^2y}{dx^2}-\frac{dy}{dx}+4x^3y=x^5$. 7

(b) Evaluate $\lim_{x\to 0}\frac{e^x-1}{\sqrt{1-\cos x}}=1$ and show that $\lim_{x\to 0}\frac{e^x-1}{x}=1$. 7

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8. (a) Find the value of 'a' if 7

$$f(x)=\begin{cases}2x-1 ; & x<2\\ a ; & x=2\\ x+1 ; & x>2\end{cases}.$$

(b) Find the length of the astroid : 7

$x^{2/3}+y^{2/3}=a^{2/3}$.

9. (a) Find the vertex, focus and directrix of the parabola : 7

$x^2+8x+12y+4=0$.

(b) Find the equation of the hyperbola whose focus is (1, 1), directrix $2x+2y=1$ and eccentricity is $\sqrt{2}$. 7

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Paper Code : 21304

F–404

B.C.A. (Ist Semester) Examination, 2019–20

MATHEMATICS–I

(New Course)

Paper No. BCA–104–N

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Evaluate :

$$\lim_{x\to 0}\frac{x^{1/2}\tan x}{(e^x-1)^{3/2}}$$

(b) Discuss the continuity of $f(x) = [x]$ at $x = 0$, where $[x]$ denotes the integral part of x.

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2. (a) Find n^{th} differential coefficient of $\tan^{-1}\frac{x}{a}$.

(b) If :

$$y^{1/m} + y^{-1/m} = 2x,$$

prove that :

$$(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)\,y_n = 0$$

3. (a) If :

$$u = \tan^{-1}\frac{xy}{\sqrt{1+x^2+y^2}},$$

show that :

$$\frac{\partial^2 u}{\partial x\partial y} = \frac{1}{(1+x^2+y^2)^{3/2}}$$

(b) If :

$$u = \sin^{-1}\frac{x^2+y^2}{x+y},$$

show that :

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$$

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4. (a) If :

$$x = a \cosh \xi \cos \eta$$

$$y = a \sinh \xi \sin \eta$$

prove that :

$$\frac{\partial(x, y)}{\partial(\xi, \eta)} = \frac{a^2}{2}(\cosh 2\xi - \cos 2\eta)$$

5. (a) Evaluate :

$$\int_0^{2a} \frac{x^{a/2}}{\sqrt{2a - x}} dx$$

(b) Show that the ordinate $x = a$ divides the area between $y^2(2a - x) = x^3$ and its asymptote into two parts in the ratio $3\pi - 8$; $3\pi + 8$.

6. (a) Prove that :

$$\nabla^2(r^n \bar{r}) = n(n+3)r^{n-2}\bar{r}$$

(b) Show that :

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7. (a) Solve :

$$(1 + x^2)\frac{dy}{dx} + y = \tan^{-1} x$$

(b) Solve :

$$(D^2 + 1)y = 8x^2 e^{2x} \sin 2x$$

8. (a) Obtain the equation of the ellipse whose letus rectum is 5 and whose eccentricity is 2/3, the axis of the ellipse being the axes of coordinates.

(b) Find the eccentricity and co-ordinate of the foci of the curve $3x^2 - y^2 = 4$.

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Paper Code : 21304

F–404

B.C.A. (First Semester) Examination, 2018

(New Course)

Paper-No. BCA-104-N

MATHEMATICS-I

Time : 3 Hours] [Maximum Marks : 70

Note :- Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Determine the values of a, b, c for which the function :

$$f(x) = \begin{cases} \dfrac{\sin(a+1)x + \sin x}{x} & : x < 0 \\ c & : x = 0 \\ \dfrac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & : x > 0 \end{cases}$$

is continuous at $x = 0$.

(b) Examine the function defined below for continuity at $x = 0$:

$$f(x) = \begin{cases} \dfrac{\sin^2 ax}{x^2} & : x \neq 0 \\ 1 & : x = 0 \end{cases}$$

2. (a) If :

$$I_n = \frac{d^n}{dx^n}(x^n \log x)$$

prove that $I_n = nI_{n-1} + (n-1)!$

(b) If $y = a\cos(\log x) + b\sin(\log x)$, show that :

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + (x^2+1)y_n = 0.$$

3. (a) If $u = e^{xyz}$.

show that :

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$$

(b) If $(\tan x)^y + y^{\cos x} = a$.

find the value of $\dfrac{dy}{dx}$.

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4. (a) If :

$$x = r \sin\theta \cos\phi$$

$$y = r \sin\theta \sin\phi$$

$$z = r \cos\theta.$$

show that $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin\theta$

(b) Evaluate :

$$\int_0^{\pi/2} \log \sin x \, dx$$

5. (a) Obtain the reduction formula for :

$$\int \sec^n x \, dx$$

(b) Find the area bounded by the parabola $y^2 = 4ax$ and its latus rectum

6. (a) Find the length of the astroid :

$$x^{2/3} + y^{2/3} = a^{2/3}$$

(b) Show that the volume of a sphere of radius a is $\frac{4}{3}\pi a^3$.

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7. (a) Evaluate grad r^n.

(b) If $r = a\cos t\,\hat{i} + a\sin t\,\hat{j} + at\tan\alpha\,\hat{k}$, find :

$$\left|\frac{dr}{dt} \times \frac{d^2r}{dt^2}\right|$$

8. (a) Solve the differential equation :

$$(D^4 + m^4)y = 0$$

(b) Solve :

$$(D^2 + a^2)y = \sec ax$$

9. (a) Find the equation of the hyperbola whose focus is (1, 1), directrix $2x + 2y = 1$ and eccentricity is $\sqrt{2}$.

(b) Find the vertex, focus and directrix of the parabola :

$$x^2 + 8x + 12y + 4 = 0$$

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Paper Code : 21304

F–404

B. C. A. (First Semester)
EXAMINATION, 2017-18
(New Course)

Paper No. BCA—104—N

MATHEMATICS—I

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Find the right hand and left hand limits of $\lim_{x \to 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}$.

   (b) Show that the function f defined as :

$$f(x) = \begin{cases} 0 & : \ x = 0 \\ \frac{1}{2} - x & : \ 0 < x < \frac{1}{2} \\ \frac{1}{2} & : \ x = \frac{1}{2} \\ \frac{3}{2} - x & : \ \frac{1}{2} < x < 1 \\ 1 & : \ x = 1 \end{cases}$$

   has three points of discontinuity which you are required to find.

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2. (a) If : $y = \sin mx + \cos mx$

   Prove that :

$$y_n = m^n \left[1 + (-1)^n \sin 2mx \right]^{1/2}$$

   (b) If : $y = e^{a \cos^{-1} x}$

   prove that :

$$(1 - x^2) y_{n+2} - (2n + 1) x y_{n+1} - (n^2 + a^2) y_n = 0$$

3. (a) If : $u = \tan^{-1} \frac{xy}{\sqrt{1 + x^2 + y^2}}$

   show that :

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{1}{(1 + x^2 + y^2)^{3/2}}$$

   (b) State and prove Euler's theorem.

4. The roots of the equation in λ :

$$(\lambda - x)^3 + (\lambda - y)^3 + (\lambda - z)^3 = 0$$

   are u, v, w.

   Prove that :

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = -2 \frac{(x - y)(y - z)(z - x)}{(u - v)(v - w)(w - u)}$$

5. (a) If : $I_n = \int_0^{\pi/4} \tan^n x \, dx$

   prove that :

$$I_{n+1} + I_{n-1} = \frac{1}{n}$$

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(b) Evaluate $\iint x^2 y^2 dx\, dy$ over that region $x^2 + y^2 \leq 1$.

6. (a) Find the whole area of the curve :

$$a^2 y^2 = x^3 (2a - x)$$

(b) Show that the volume of a sphere of radius a is $\frac{4}{3}\pi a^3$.

7. (a) Show that :

$$\hat{r} \times d\hat{r} = \frac{\hat{r} \times d\hat{r}}{r^2}$$

(b) Prove that :

$$\text{div grad } r^n = n(n+1) r^{n-2}$$

8. (a) Solve :

$$\frac{dy}{dx} + 2y \tan x = \sin x$$

given that $y = 0$ when $x = \frac{\pi}{3}$.

(b) Solve :

$$(D^2 - 2D + 1) y = x e^x \sin x$$

9. (a) Discuss the properties of parabola and hyperbola.

(b) Trace the ellipse :

$$3x^2 + 4y^2 = 12$$

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Paper Code : 21304

F–404

B. C. A. (First Semester)
EXAMINATION, 2016-17
(New Course)

Paper No. BCA–104–N

MATHEMATICS-I

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Show that $\lim_{x \to 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^3}}$ does not exist.

   (b) Discuss the types of discontinuities.

2. (a) A function $f(x)$ is defined as :

$$f(x) = \begin{cases} \frac{x^2}{a} - a & : \ x < a \\ 0 & : \ x = a \\ a - \frac{a^2}{x} & : \ x > a \end{cases}$$

   Show that the function $f(x)$ is continuous at $x = a$.

   (b) Find nth derivative of $\tan^{-1} \frac{x}{a}$ with respect to x.

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3. (a) If $x = \sin \left\{ \frac{(\log y)}{a} \right\}$, prove that :

$$(1 - x^2) y_{n+2} - (2n + 1) x y_{n+1} - (n^2 + a^2) y_n = 0$$

   (b) If $u = e^{xyz}$ show that :

$$\frac{\partial^3 u}{\partial x \, \partial y \, \partial z} = e^{xyz} \{ 1 + 3xyz + x^2 y^2 z^2 \}$$

4. (a) If $x = a \cosh \xi \cos \eta$,

$y = a \sinh \xi \sin \eta$

   prove that :

$$\frac{\partial (x, y)}{\partial (\xi, \eta)} = \frac{a^2}{2} (\cosh 2\xi - \cos 2\eta)$$

   (b) If $I_n = \int_0^{\frac{\pi}{3}} \tan^n x \, dx$, show that :

$$(n - 1)(I_n + I_{n-2}) = \left(\sqrt{3} \right)^{n-1}$$

5. (a) Evaluate $\int_0^{\frac{\pi}{2}} \log \sin x \, dx$

   (b) Find the entire area between the curve $y^2 (2a - x) = x^3$ and its asymptote.

6. (a) Solve :

$$\frac{x \, dx + y \, dy}{x \, dy - y \, dx} = \sqrt{\frac{a^2 - x^2 - y^2}{x^2 + y^2}}$$

   (b) Solve :

$$(D^2 + D + 1) y = \sin 2x$$

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7. (a) Show that :

$$\bar{\nabla} r^n = n r^{n-2} \bar{r}$$

(b) Show that the vectors $\bar{a} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\bar{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\bar{c} = 2\hat{i} + \hat{j} - 4\hat{k}$ form a right angled triangle.

8. (a) Find the centre, vertices, foci, eccentricity and letus rectum of the hyperbola :

$$9x^2 - 16y^2 - 18x - 64y - 199 = 0$$

(b) One extremity of focal chord of the parabola $y^2 = 16x$ is the point (1, 4), find the co-ordinates of the other extremity.

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